5 Dependenycy Injection

Dependency Injection (DI) is a design pattern used to reduce tight coupling between components in software. Laravel fully embraces DI and uses it extensively through its **Service Container**—a powerful tool for managing class dependencies and performing dependency resolution automatically.

5.1 What is Dependency Injection?

Dependency Injection means that instead of a class creating its dependencies, those dependencies are “injected” from the outside—usually via the constructor or method parameters.

This allows for more flexible, testable, and maintainable code.

For demo, let’s say we have a `ReportService` that uses a `FileLogger`:

```
class ReportService {
    public function __construct() {
        $this->logger = new FileLogger();
    }
}
```

This creates a tight coupling between `ReportService` and `FileLogger`. If we want to change the logger to `DatabaseLogger`, we have to modify the `ReportService` class.

```
class ReportService {
    public function __construct() {
        $this->logger = new DatabaseLogger();
    }
}
```

This is not ideal for testing or maintaining the code.

Now, let’s see how we can use **Dependency Injection** to improve this.

```
class ReportService {
    public function __construct(FileLogger $logger) {
        $this->logger = $logger;
    }
}
```

This approach makes code more **modular**, **testable**, and **maintainable**.